

A stabilized finite element method for enforcing stiff anisotropic cohesive laws using interface elements

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Received 23 October 2018; received in revised form 27 January 2019; accepted 9 February 2019

Available online 19 February 2019

Abstract

We present a stabilized finite element method that generalizes Nitsche's method for enforcing stiff anisotropic cohesive laws with different normal and tangential stiffness. For smaller values of cohesive stiffness, the stabilized method resembles the standard method, wherein the traction on the crack surface is enforced as a Neumann boundary condition. Conversely, for larger values of cohesive stiffness, the stabilized method resembles Nitsche's method, wherein the cohesive law is enforced as a kinematic constraint. We present several numerical examples, in two-dimensions, to compare the performance of the stabilized and standard methods. Our results illustrate that the stabilized method enables accurate recovery of crack-face tractions for stiff isotropic and anisotropic cohesive laws; whereas, the standard method is less accurate due to spurious traction oscillations. Also, the stabilized method could mitigate spurious sensitivity of load–displacement results to displacement increment in mixed-mode fracture simulation, owing to its stability and accuracy.

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Keywords: Interface elements; Nitsche's method; Traction oscillations; Numerical stability; Cohesive zone models; Mixed-mode fracture

1. Introduction

Numerical simulation of fracture propagation in multi-dimensions is a challenging problem, in part due to mixed-mode interactions and stiff anisotropic cohesive response at arbitrarily shaped crack interfaces, particularly under compression. The cohesive zone modeling approach has been widely used to analyze and predict mixed-mode fracture or delamination propagation at anisotropic and/or dissimilar material interfaces, despite its limitations. Typically, cohesive zone models (CZMs) are implemented in conjunction with the finite element method by introducing zero-thickness interface elements along potential crack surfaces. The constitutive behavior of the interface is defined by a cohesive law that relates the traction with the separation across the crack surface. There are broadly two classes of CZMs: intrinsic with initially elastic cohesive laws and extrinsic with initially rigid cohesive laws. Cohesive laws with large initial stiffness may be required for accurate fracture analysis and contact enforcement in the intrinsic approach; however, stiff cohesive laws may also need to be enforced during cycle-by-cycle fatigue analysis even in the extrinsic approach (see Fig. 1). It is well-known that the standard (penalty-like)

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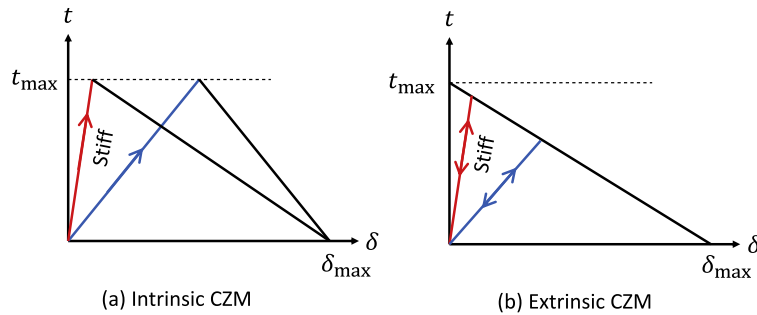


Fig. 1. Illustration of stiff (red line) cohesive laws encountered in fracture analysis under monotonic and cyclic loading. (a) In intrinsic CZMs, stiff cohesive laws may be prescribed to define the linear elastic portion before damage initiation or no-interpenetration (contact) condition; (b) In extrinsic CZMs, stiff cohesive laws may be encountered during unloading/reloading immediately after damage initiation or contact enforcement during cyclic loading. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

method for enforcing stiff cohesive laws using interface elements suffers from a distinct numerical instability that is often manifested by spurious oscillations in crack-face tractions, which may cause inaccuracies and convergence issues. The purpose of this paper is to introduce a novel stabilized finite element method by generalizing Nitsche's method for enforcing stiff anisotropic cohesive laws that alleviates the numerical instability issue afflicting the standard method.

In an extrinsic CZM, it is assumed that the interface separates only after the cohesive traction exceeds a finite cohesive strength and then the maximum cohesive traction decreases monotonically with the increase in separation until an ultimate separation value is reached (see Fig. 1b). Since their introduction in the seminal paper by Camacho and Ortiz [1], the extrinsic CZM approach has been used to simulate dynamic fracture in a wide-range of engineering materials [e.g., 2–7]. In practice, the extrinsic approach is implemented by adaptively inserting zero-thickness interface elements in a finite element mesh adjacent to the crack tip when a specified criterion for the onset of failure is met. Consequently, the numerical implementation of extrinsic CZMs is computationally efficient, but it requires sophisticated topology change algorithms to modify the associated finite element data structures consistently with the evolving fracture geometry, which increases algorithmic complexity. Despite several algorithmic advances [8–11], numerical implementation of extrinsic CZMs in a legacy finite element framework remains non-trivial, compared to intrinsic CZMs that are much more straightforward to implement.

In intrinsic CZMs, it is assumed that the cohesive traction increases gradually with separation till it reaches a finite cohesive strength in the linear elastic regime, and then it decreases monotonically till the separation reaches an ultimate value, where complete de-cohesion occurs (see Fig. 1a). Xu and Needleman [12] pioneered the intrinsic approach using a potential-based traction–separation law to model dynamic fracture growth in brittle solids. Several phenomenological and potential-based intrinsic CZMs with bilinear [13], trapezoidal [14,15], and polynomial shapes [16,17] have been developed for specific applications. In practice, the intrinsic approach is implemented by inserting zero-thickness interface elements in a finite element mesh along all potential crack paths *a priori*. This approach has been extensively used to model both quasi-static and dynamic failure in a variety of applications including particle–matrix interface debonding in metal-based composites [18], delamination in laminated composite materials [19–21], and fiber–metal laminate failure [22]. The intrinsic approach is relatively straightforward to implement in a legacy finite element framework, but it has some shortcomings, including the well-known “artificial compliance” [23,24] and increased computational cost.

Despite their differences, the implementation of both extrinsic and intrinsic approaches for stiff cohesive laws using the standard (penalty-like) method is prone to several numerical issues during dynamic and quasi-static fracture analysis. The stiffness of the cohesive law is often defined using a non-dimensional quantity $M = \alpha^0 h / E$, where α^0 is the initial cohesive stiffness, E is the elastic modulus and h is the mesh size parameter [25]. For laminated composites, Turon et al. [26] suggested using the sub-laminate thickness t , instead of mesh size parameter h , to define the non-dimensional quantity M . For a given size-scale parameter (i.e., h or t), choosing a large cohesive stiffness α^0 relative to the elastic modulus E leads to a stiff cohesive law with $M > 1000$. In dynamic fracture analysis, to avoid the artificial compliance issue, it is required to assume an adequately large value for the initial

cohesive stiffness in intrinsic CZMs [23,25,27], but that can lead to ill-conditioning of the tangent stiffness matrices. This ill-conditioning issue may also arise in extrinsic CZMs under cyclic loading, if the interface is unloaded immediately after crack initiation when the elastic unloading/reloading slope could be large [1]. This issue can be resolved to some extent by restricting the time step in an explicit finite element scheme to an extremely small value [4,24], but this will result in an impractically high computation cost. In quasi-static fracture analysis, intrinsic CZMs exhibit spurious traction oscillations along the cohesive interface, especially near crack tips, if a large initial cohesive stiffness is specified [26,28–30], or if the cohesive interface has a curved geometry and its behavior under compressive loading is described by an anisotropic cohesive law with different values for normal and tangential stiffness [31]. It has been argued in [31] that the issue of spurious traction oscillations encountered in the standard (penalty-like) method for intrinsic CZMs with stiff cohesive laws (using full or reduced integration) arises due to the violation of the inf-sup or LBB condition.

In the traditional penalty method [32], a Dirichlet constraint at the interface is enforced by introducing a spring-like tie constraint at the interface. A better approximation of the Dirichlet constraint can be obtained by using a large value for the penalty parameter, which can be interpreted as the stiffness of the spring (i.e., initial slope of the traction–separation law). Thus, the standard finite element method for implementing an intrinsic CZM is equivalent to the penalty method for stiff cohesive laws. Theoretically, the interfacial constraint of zero separation before crack initiation (i.e., extrinsic CZM) can be achieved if the initial cohesive stiffness approaches infinity, but using a very large cohesive stiffness leads to ill-conditioning and numerical instability issues. Lagrange-multiplier-based mixed formulations can alleviate instability issues associated with cohesive interface elements [33,34] or with embedded contact interfaces in the extended finite element method (XFEM) [35,36]. However, these approaches can be computationally costly and complicated to implement, because it is difficult and non-trivial to find a stable Lagrange multiplier space that alleviates traction oscillations [37]. Another alternative is Nitsche’s method, which was originally introduced in [38] to weakly enforce Dirichlet boundary conditions. Later, it was extended to weakly enforce the continuity of the displacement field at the interior boundaries [39]. We note that the discontinuous Galerkin (DG) method essentially originated from Nitsche’s method [40] and the latter has been referred to as the classical DG method [41]. A comprehensive review of Nitsche’s method and its application to interface problems can be found in [42,43].

The Nitsche’s method can be interpreted as a variationally consistent penalty method for weakly enforcing interfacial constraints. This method can eliminate the instability issues associated with the penalty method by adding consistency terms [44,45], and can yield oscillation-free traction profiles at embedded interfaces [46]. In the recent decades, Nitsche’s method has been utilized for solving a wide range of interface problems in an efficient way [46–55]. More recently, Nitsche-based methods have been developed for frictional-sliding on embedded interfaces [56,57] and small-sliding contact on frictional surfaces, including stick–slip behavior [58]. In this article, we propose a stabilized finite element method for cohesive fracture problems, which is inspired by the Nitsche’s method for general boundary conditions developed by Juntunen and Stenberg [59]. The proposed method ensures accurate recovery of crack-surface traction even for large values of cohesive stiffness (e.g., 8–16 orders of magnitude more than bulk stiffness). We further demonstrate the ability of the stabilized method to alleviate numerical instability associated with the implementation of stiff, anisotropic cohesive laws with different interface properties in the normal and tangential directions. The main novelty of this paper is that it extends Nitsche’s method for cohesive fracture so that it is applicable to both intrinsic and extrinsic approaches, including stiff elastic loading and unloading conditions. The rest of this paper is organized as follows: in Section 2, we introduce the governing equations of the cohesive fracture problem and the weak forms corresponding to the standard and stabilized methods; in Section 3, we discuss the numerical implementation of the stabilized method in the commercial finite element software Abaqus; in Section 4, we present several numerical examples to compare the standard and stabilized methods with a particular emphasis on the accuracy of evaluating crack-face tractions for stiff cohesive laws in quasi-static simulation, including mixed-mode fracture; in Section 5, we conclude with a summary and closing remarks.

2. Model formulation

In this section, we present details of the stabilized finite element method for enforcing stiff cohesive laws. We first present the strong form of the governing equations followed by a brief description of the interface cohesive law for mixed-mode loading. We next derive the weak form for the standard (penalty-like) and stabilized (Nitsche-inspired) methods.

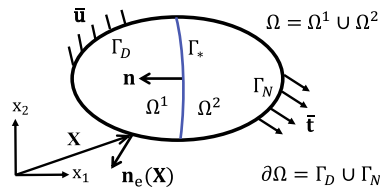


Fig. 2. A schematic of the domain for the quasi-static cohesive fracture problem.

2.1. Strong form

We consider the initial domain $\Omega \subset \mathbb{R}^{nd}$ containing a linearly elastic solid with $nd = 2$ in two dimensions, as shown in Fig. 2. The domain boundary $\Gamma \equiv \partial\Omega$ is partitioned into two disjoint parts such that $\partial\Omega = \Gamma_D \cup \Gamma_N$ with $\Gamma_D \cap \Gamma_N = \emptyset$, where the Dirichlet and Neumann boundary conditions are enforced. The domain Ω contains an internal cohesive interface Γ_* , which divides Ω into two non-overlapping sub-domains Ω^1 and Ω^2 . The outward unit normal to the boundary $\partial\Omega$ is denoted by $\mathbf{n}_e$, and the unit normal vector associated with the interface Γ_* denoted by $\mathbf{n}$ points from Ω^2 to Ω^1 . We use a rectangular Cartesian coordinate system and the total Lagrangian description for variables with $\mathbf{X}$ denoting the position of the material points.

The displacement field $\mathbf{u}(\mathbf{X})$ is discontinuous across Γ_* , but continuous in Ω^1 and Ω^2 ; therefore, it can be represented by two continuous functions $\mathbf{u}^1(\mathbf{X})$ and $\mathbf{u}^2(\mathbf{X})$ in the respective sub-domains. For brevity, henceforth we will suppress the spatial dependence of variables. Assuming small displacements, the Cauchy stress tensor can be defined in Ω^1 and Ω^2 as

$$\boldsymbol{\sigma}^m = \mathbf{D}^m : \boldsymbol{\epsilon}^m, \quad m = \{1, 2\}, \quad (1)$$

where $\mathbf{D}$ denotes the fourth-order elasticity tensor and the small strain tensor $\boldsymbol{\epsilon} = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is defined by the symmetric part of the displacement gradient tensor and ∇ is the spatial gradient operator with respect to the material coordinates $\mathbf{X}$.

The strong form of the quasi-static boundary value problem in the absence of body force is:

$$\nabla \cdot \boldsymbol{\sigma}^m = \mathbf{0} \quad \text{in } \Omega^m, \quad m = \{1, 2\}, \quad (2)$$

$$\mathbf{u} = \bar{\mathbf{u}} \quad \text{on } \Gamma_D, \quad (3)$$

$$\boldsymbol{\sigma} \cdot \mathbf{n}_e = \bar{\mathbf{t}} \quad \text{on } \Gamma_N, \quad (4)$$

$$\mathbf{t}_c(\delta) = \boldsymbol{\sigma} \cdot \mathbf{n}, \quad \text{on } \Gamma_*, \quad (5)$$

where $\bar{\mathbf{t}}$ is the prescribed traction or stress vector on the Neumann boundary Γ_N , $\bar{\mathbf{u}}$ is the prescribed displacement vector on the Dirichlet boundary Γ_D , and the traction on the cohesive interface $\mathbf{t}_c$ is given by a function of the interface separation or displacement jump as

$$\delta = \llbracket \mathbf{u} \rrbracket = \mathbf{u}^2 - \mathbf{u}^1 \quad (6)$$

Note that the traction $\mathbf{t}_c$ is continuous across the cohesive interface Γ_* and is related to the Cauchy stress tensor evaluated in the sub-domains Ω^1 and Ω^2 as

$$\mathbf{t}_c = \boldsymbol{\sigma} \cdot \mathbf{n} = -\boldsymbol{\sigma}^1 \cdot \mathbf{n}^1 = \boldsymbol{\sigma}^2 \cdot \mathbf{n}^2. \quad (7)$$

In the above two equations, we followed the notation convention used in [52] for defining the displacement jump and outward normal, and use it to establish the weak form in Section 2.3.

2.2. Intrinsic cohesive law

For simplicity, we consider the bilinear intrinsic cohesive law with an initial (increasing) linear elastic portion followed by a (decreasing) linear softening response. The corresponding relation between crack-face traction and interface separation can be defined using the damage mechanics framework as [20,21,61]

$$\mathbf{t}_c = -\alpha \delta, \quad (8)$$

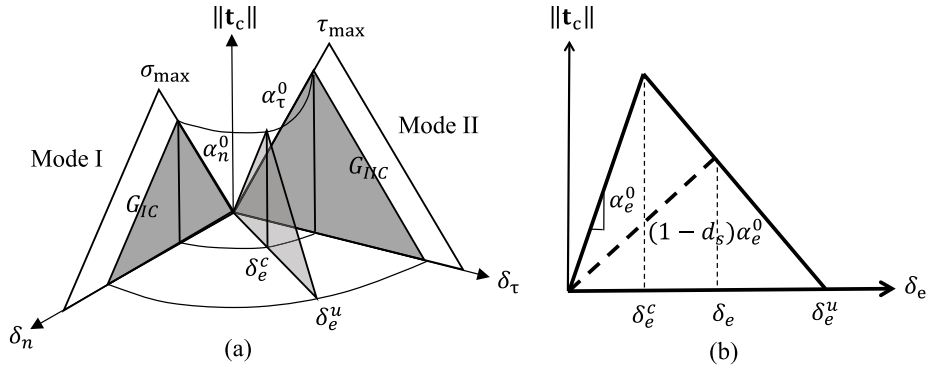


Fig. 3. A schematic diagram of the mixed-mode bilinear cohesive law (redrawn from [60]): (a) the traction–separation relationship for any arbitrary mode-mix ratio is defined in terms of the pure mode I and mode II relationships; (b) the relationship between the static damage variable d_s and the equivalent separation. The magnitude of the traction vector $\|\mathbf{t}_c\| = \sqrt{t_n^2 + t_\tau^2}$.

where α is the cohesive stiffness matrix including the effect of damage. Note that the crack-face traction $\mathbf{t}_c$ is the Newton's third law pair to the cohesive traction. Thus, the negative sign in the above equation indicates that $\mathbf{t}_c$ causes a restoring force at the interface, which is equal and opposite to the deforming force. To represent the mixed mode-I and mode-II fracture behavior in two dimensions, we use the normal and tangential coordinate system. Accordingly, the tangential t_τ and normal t_n components of the traction vector $\mathbf{t}_c$ are related to the tangential δ_τ and normal δ_n components of the interface separation δ as

$$\begin{Bmatrix} t_\tau \\ t_n \end{Bmatrix} = - \begin{bmatrix} (1 - d_s)\alpha_\tau^0 & 0 \\ 0 & (1 - d_s)\alpha_n^0 \end{bmatrix} \begin{Bmatrix} \delta_\tau \\ \delta_n \end{Bmatrix}, \quad (9)$$

where α_n^0 and α_τ^0 represent the initial cohesive stiffness in the normal and the tangential directions, respectively. The scalar damage variable d_s describing interface degradation under quasi-static mixed-mode loading in two dimensions is given by

$$d_s = \begin{cases} 0 & \text{if } \delta_e < \delta_e^c, \\ \frac{\delta_e^u(\delta_e - \delta_e^c)}{\delta_e(\delta_e^u - \delta_e^c)} & \text{if } \delta_e^c \leq \delta_e < \delta_e^u, \\ 1 & \text{if } \delta_e^u \leq \delta_e, \end{cases} \quad (10)$$

where $\delta_e = \sqrt{\delta_n^2 + \delta_\tau^2}$ is the equivalent separation, δ_e^c and δ_e^u are interface parameters corresponding to critical and ultimate separations, respectively, defined as [62]

$$\frac{1}{\delta_e^c} = \sqrt{\left(\frac{\alpha_n^0 \cos I}{\sigma_{\max}}\right)^2 + \left(\frac{\alpha_\tau^0 \cos II}{\tau_{\max}}\right)^2} \quad (11)$$

$$\frac{1}{\delta_e^u} = \left(\frac{\alpha_n^0 \delta_e^c (\cos I)^2}{2 G_{IC}}\right) + \left(\frac{\alpha_\tau^0 \delta_e^c (\cos II)^2}{2 G_{IIC}}\right) \quad (12)$$

where the direction cosines $\cos I = \delta_n/\delta_e$ and $\cos II = \delta_\tau/\delta_e$, $\sigma_{\max}$ and $\tau_{\max}$ are the pure mode I and mode II cohesive strengths, and G_{IC} and G_{IIC} are the pure mode I and mode II critical fracture energies (see Fig. 3a). For monotonic loading, when the equivalent interface separation δ_e is less than the critical separation δ_e^c , there is no damage in the cohesive interface elements. After the critical separation is exceeded, damage starts to accumulate till the separation reaches the ultimate value δ_e^u , when the cohesive elements are completely damaged (see Fig. 3b). The mixed-mode cohesive law described above was previously proposed by Jiang et al. [62], wherein quadratic damage initiation and mixed-mode failure criteria were used to obtain the equivalent critical and ultimate separations. If the parameter values of cohesive stiffness, cohesive strength and fracture energy are chosen to be the same for both normal and shear modes, then we get an isotropic cohesive law, else we get an anisotropic cohesive law. For non-monotonic (cyclic) loading, we can enforce irreversibility of damage evolution by ensuring that damage does not change during unloading and reloading cycles, until the previous maximum damage is exceeded [61,63].

2.3. Weak form

We follow the Galerkin method of weighted residuals to derive the weak forms corresponding to the standard and stabilized methods. We define the space of trial functions $\mathbf{U}$ and the space of test functions $\mathbf{W}$, such that:

$$\mathbf{U} = \{\mathbf{u} \in H^1(\Omega), \mathbf{u} = \bar{\mathbf{u}} \text{ on } \Gamma_D\}, \quad (13)$$

$$\mathbf{W} = \{\mathbf{w} \in H^1(\Omega), \mathbf{w} = \mathbf{0} \text{ on } \Gamma_D\}. \quad (14)$$

By weighting Eq. (2) with the test function $\mathbf{w}$, integrating by parts, applying the divergence theorem, and using the traction continuity condition at the interface in Eq. (7), and the constitutive relation in Eq. (1), we can derive the weak form as follows:

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega - \int_{\Gamma_*} (\mathbf{w}^2 \cdot \boldsymbol{\sigma}^2 \cdot \mathbf{n}^2 + \mathbf{w}^1 \cdot \boldsymbol{\sigma}^1 \cdot \mathbf{n}^1) \, d\Gamma - \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma = 0, \quad (15)$$

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega - \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \mathbf{t}_c \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma, \quad (16)$$

Note that in Eq. (15) we considered the integrals on the two sides of the cohesive interface, separately, and in Eq. (16) we defined the jump in the test function as $\llbracket \mathbf{w} \rrbracket = \mathbf{w}^2 - \mathbf{w}^1$.

2.3.1. Standard method

Substituting the traction–separation relation in Eq. (8) into the weak form in Eq. (16) we get

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega + \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \boldsymbol{\alpha} \, \delta \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma. \quad (17)$$

Thus, in the standard method the cohesive tractions are enforced as a Neumann boundary condition on the interface. Because the cohesive tractions and separations are defined in the normal and tangential directions, the weak form is implemented as,

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega + \int_{\Gamma_*} (1 - d_s) (\llbracket w_n \rrbracket \alpha_n^0 \delta_n + \llbracket w_\tau \rrbracket \alpha_\tau^0 \delta_\tau) \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma. \quad (18)$$

If the initial cohesive stiffness parameters α_n^0 and α_τ^0 are taken to be sufficiently large the standard method resembles the penalty method for enforcing displacement continuity across the interface. However, for stiff cohesive laws, that is, if cohesive stiffness is several orders of magnitude greater than the elastic modulus, the standard method becomes ill-conditioned leading to instability and/or convergence issues. In the limiting case where $\alpha_n^0 \rightarrow \infty$ and/or $\alpha_\tau^0 \rightarrow \infty$, that is, for a non-interpenetration (contact) constraint or an extrinsic cohesive law, the standard method is not well defined. To circumvent the above issues, discontinuous Galerkin or Nitsche-based methods have been proposed in [55,64–67] based on the extrinsic approach, wherein the interface is perfectly bonded until a certain stress threshold (i.e., interface separation tends to zero and cohesive stiffness tends to infinity). While in [64] interface bonding prior to separation was enforced using the interior penalty method, in [66] Riemann solutions were used to enforce interface conditions. In the following section, we will present an alternative stabilized finite element method for cohesive fracture that is applicable for the whole range of values of cohesive stiffness $\alpha_n, \alpha_\tau > 0$, so that it is applicable to both intrinsic and extrinsic approaches.

2.3.2. Stabilized method

The proposed method adopts the approach developed in [52,59] and generalizes it to cohesive fracture problems. By multiplying both sides of Eq. (8) with a stabilization matrix $\mathbf{S}$ we obtain

$$\mathbf{S} \mathbf{t}_c = -\mathbf{S} \boldsymbol{\alpha} \, \delta. \quad (19)$$

After multiplying the above equation by the weighting function $\mathbf{w}$ and integrating over the cohesive interface Γ_* we get

$$\int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \mathbf{S} \mathbf{t}_c \, d\Gamma = - \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \mathbf{S} \boldsymbol{\alpha} \, \delta \, d\Gamma. \quad (20)$$

By adding the above equation to the weak form in Eq. (16), we obtain

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega - \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot (\mathbf{I} - \mathbf{S}) \mathbf{t}_c \, d\Gamma + \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \mathbf{S} \boldsymbol{\alpha} \, \delta \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma, \quad (21)$$

where $\mathbf{I}$ is the identity matrix. The interface traction can be defined as

$$\mathbf{t}_c = \langle \boldsymbol{\sigma} \rangle_{\gamma} \cdot \mathbf{n} \quad \text{on } \Gamma_*, \quad (22)$$

where the weighted average of the stress tensors on both sides of the interface is given by

$$\langle \boldsymbol{\sigma} \rangle_{\gamma} = (\gamma^1 \boldsymbol{\sigma}^1 + \gamma^2 \boldsymbol{\sigma}^2) \quad \forall \quad \gamma^1 + \gamma^2 = 1, \quad \gamma^1 > 0, \quad \gamma^2 > 0. \quad (23)$$

Choosing the weights $\gamma^1 = \gamma^2 = 0.5$, yields the mean of the stress tensors evaluated on both sides of the interface. To complete the formulation, we define the stabilization matrix $\mathbf{S}$ as

$$\mathbf{S} = \begin{bmatrix} \frac{\beta_{\tau}}{\alpha_{\tau}^0(1-d_s) + \beta_{\tau}} & 0 \\ 0 & \frac{\beta_n}{\alpha_n^0(1-d_s) + \beta_n} \end{bmatrix}, \quad (24)$$

where β_{τ}, β_n are the stabilization parameters. The stabilization parameters β_{τ}, β_n and the weights γ^1, γ^2 play a key role in the numerical performance of the method. This so-called weighted Nitsche method [58] is particularly advantageous for dissimilar material interfaces with large contrast in material properties or for unstructured meshes with significant variations in mesh size. For constant strain triangular (CST) and tetrahedral elements, Annavarapu et al. [46] provided estimates for the stabilization parameters using a local coercivity analysis as given by

$$\beta_n = \beta_{\tau} = 2 \left(\frac{|\mathbf{D}^1|(\gamma^1)^2}{\text{meas}(\Omega^1)} + \frac{|\mathbf{D}^2|(\gamma^2)^2}{\text{meas}(\Omega^2)} \right) \text{meas}(\Gamma_*) \quad (25)$$

where $|\mathbf{D}|$ denotes the two-norm of the elasticity tensor, $\text{meas}(\Omega)$ denotes the area of neighboring bulk element in 2D, and $\text{meas}(\Gamma_*)$ is the length of the interface element. With a judicious choice of the weights γ^1, γ^2 , the stabilization parameters β_{τ}, β_n scale as $1/h$, where $h \approx \text{meas}(\Gamma_*)$ is the mesh/element size parameter. For all $h \in (0, \infty)$, both the initial cohesive stiffness α and the stabilization parameter β scale as $1/h$; thus, the stabilized method provides a well-conditioned discrete system, irrespective of the mesh size, by ensuring that the cohesive stiffness terms and the bulk stiffness terms have the same scaling. For an elaborate discussion on the appropriate choice of weights, we refer the reader to Ref. [46]. Note that, in this study, we used the estimates given in (25) to calculate the stabilization parameters for bilinear quadrilateral elements; precise estimates can be derived as in [46], but such analysis is beyond the scope of this paper.

Finally, the weak form for the stabilized method can be written as

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega - \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot (\mathbf{I} - \mathbf{S}) \langle \boldsymbol{\sigma} \rangle_{\gamma} \cdot \mathbf{n} \, d\Gamma + \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \mathbf{S} \boldsymbol{\alpha} \, \delta \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma. \quad (26)$$

In the above equation, the second and third terms on the left hand side ensure consistency and stability of the proposed method, respectively. The stabilized method presented here is unsymmetric and resembles the incomplete interior penalty method [43,68]. It can be proved that the displacement solution $\mathbf{u}$ of the strong form equations (2) – (5) is satisfied by the solution to the weak form equation (26), which establishes consistency for any value of cohesive stiffness; the mathematical procedure for proving this is similar to that described in [59, Lemma 2.1]. As $(1-d_s)\alpha_n^0, (1-d_s)\alpha_{\tau}^0 \rightarrow \infty$ (refer to Eq. (24)), we recover the Nitsche-based method for frictional contact as [56]

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega - \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot \langle \boldsymbol{\sigma} \rangle_{\gamma} \cdot \mathbf{n} \, d\Gamma + \int_{\Gamma_*} \llbracket \mathbf{w} \rrbracket \cdot (\beta_{\tau} \llbracket u_{\tau} \rrbracket + \beta_n \llbracket u_n \rrbracket) \, d\Gamma = \int_{\Gamma_N} \mathbf{w} \cdot \bar{\mathbf{t}} \, d\Gamma. \quad (27)$$

As $(1-d_s)\alpha_n^0, (1-d_s)\alpha_{\tau}^0 \rightarrow 0$, we recover the weak form for a traction-free crack surface as

$$\int_{\Omega} \nabla^s \mathbf{w} : \mathbf{D} : \nabla^s \mathbf{u} \, d\Omega = \int_{\Gamma_N} \delta \mathbf{u} \cdot \bar{\mathbf{t}} \, d\Gamma. \quad (28)$$

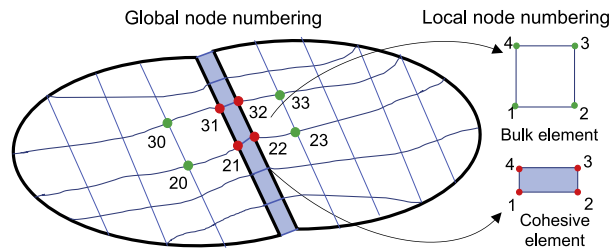


Fig. 4. Finite element discretization with bulk and zero-thickness cohesive elements.

Thus, the stabilized method remains well-defined for any arbitrary values of the cohesive stiffness terms, that is, for $(1 - d_s)\alpha_n^0, (1 - d_s)\alpha_t^0 \in [0, \infty)$. Comparing the weak forms in Eqs. (16) and (26), we can obtain an alternative definition for crack-surface traction as

$$\mathbf{t}_c = (\mathbf{I} - \mathbf{S}) \langle \boldsymbol{\sigma} \rangle_\gamma \cdot \mathbf{n} - \mathbf{S} \boldsymbol{\alpha} \delta. \quad (29)$$

Thus, the key idea of the Nitsche-inspired stabilized method for cohesive fracture is to evaluate the crack surface traction in terms of the weighted average stress in the bulk material across the interface and the traction in the cohesive interface.

3. Numerical implementation

In this section, we discuss the finite element approximation using matrix notation along with the expression for the residual and tangent matrices for the bulk and interface elements. We also present algorithms for implementing this stabilized method into the commercial software Abaqus using user defined subroutines for two-dimensional plane strain analysis.

3.1. Finite element approximation

The sub-domains Ω^1 and Ω^2 are discretized by four-noded plane strain quadrilateral bulk elements and zero-thickness four-noded interface elements are introduced at the cohesive interface Γ_* (Fig. 4). The displacement field $\mathbf{u}_{2 \times 1}^m$ at any point $\mathbf{X}$ can be approximated as

$$\mathbf{u}^m(\mathbf{X}) = \mathbf{N}(\mathbf{X}) \bar{\mathbf{u}}^m, \quad m = 1, 2, \quad (30)$$

where $\bar{\mathbf{u}}^m = [u_1^1, u_2^1, \dots, u_1^4, u_2^4]_{8 \times 1}^T$ is the nodal displacement vector for a bulk element in the subdomain Ω^m , $\mathbf{N}$ is the element shape function matrix given by

$$\mathbf{N} = \begin{bmatrix} N^1 & 0 & N^2 & 0 & N^3 & 0 & N^4 & 0 \\ 0 & N^1 & 0 & N^2 & 0 & N^3 & 0 & N^4 \end{bmatrix}_{2 \times 8}, \quad (31)$$

and N^J ($J = 1, 2, 3, 4$) are the standard finite element shape functions for the four-noded quadrilateral element. Using Voigt notation, the small-strain strain tensor $\boldsymbol{\epsilon}^m = [\epsilon_{11}, \epsilon_{22}, \epsilon_{12}]_{3 \times 1}^T$ in the bulk element can be approximated as

$$\boldsymbol{\epsilon}^m = \mathbf{B} \bar{\mathbf{u}}^m, \quad m = 1, 2, \quad (32)$$

where the strain–displacement relationship matrix is defined as

$$\mathbf{B} = \begin{bmatrix} N_{,1}^1 & 0 & N_{,1}^2 & 0 & N_{,1}^3 & 0 & N_{,1}^4 & 0 \\ 0 & N_{,2}^1 & 0 & N_{,2}^2 & 0 & N_{,2}^3 & 0 & N_{,2}^4 \\ N_{,2}^1 & N_{,1}^1 & N_{,2}^2 & N_{,1}^2 & N_{,2}^3 & N_{,1}^3 & N_{,2}^4 & N_{,1}^4 \end{bmatrix}_{3 \times 8}, \quad (33)$$

and $N_{,i}^J$ denotes the derivative of the shape function N^J with respect to material coordinate X_i ($i = 1, 2$) in two dimensions. The normal and tangential components of the displacement jump or separation across the interface

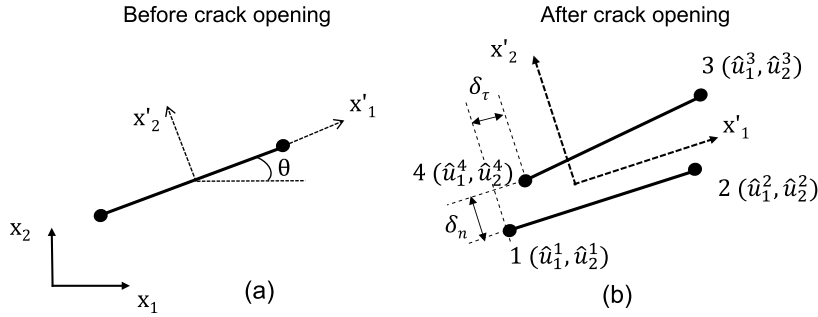


Fig. 5. Four-noded linear cohesive element: (a) the orientation θ of the local (normal-tangential) coordinates with respect to the global Cartesian coordinates before crack opening; (b) the relation between nodal displacements in global coordinates and interface separations in local coordinates after crack opening.

element can be approximated as

$$[\mathbf{u}]_{2 \times 1} = \delta_{2 \times 1} = \begin{Bmatrix} \delta_\tau \\ \delta_n \end{Bmatrix} = [\mathbf{N}] \hat{\mathbf{u}}, \quad (34)$$

where $\hat{\mathbf{u}} = [\hat{u}_1^1, \hat{u}_2^1, \dots, \hat{u}_1^4, \hat{u}_2^4]_{8 \times 1}$ is the nodal displacement vector for the four-noded interface element in Γ_* and $[\mathbf{N}]$ is the jump in the interfacial shape function matrix given by [69]

$$[\mathbf{N}] = \begin{bmatrix} -C\hat{N}^1 & -S\hat{N}^1 & -C\hat{N}^2 & -S\hat{N}^2 & C\hat{N}^2 & S\hat{N}^2 & C\hat{N}^1 & S\hat{N}^1 \\ S\hat{N}^1 & -C\hat{N}^1 & S\hat{N}^2 & -C\hat{N}^2 & -S\hat{N}^2 & C\hat{N}^2 & -S\hat{N}^1 & C\hat{N}^1 \end{bmatrix}_{2 \times 8}, \quad (35)$$

and $\hat{N}^J$ ($J = 1, 2$) are the standard finite element shape functions for the one-dimensional linear element, and C and S represent the cosine and sine, respectively, of the angle θ that defines the orientation of the interface element with the global x_1 coordinate axis (see Fig. 5). All the finite element shape functions are evaluated using the standard isoparametric concept.

3.2. Discretization and linearization

By introducing the finite element approximation into the variational form in Eq. (26), we write the discretized form of the equilibrium equations as

$$\mathbf{R}(\mathbf{U}) = \mathbf{f}_{\text{ext}} - (\mathbf{f}_{\text{int}}^b(\mathbf{U}) + \mathbf{f}_{\text{int}}^c(\mathbf{U})) = \mathbf{0}, \quad (36)$$

where $\mathbf{R}$ is the global residual vector, $\mathbf{U}$ is the global displacement vector, $\mathbf{f}_{\text{ext}}$ and $\mathbf{f}_{\text{int}}^b$ are the global external and internal force vectors, respectively, assembled from all the bulk elements in the domain Ω , and $\mathbf{f}_{\text{int}}^c$ is the internal force vector from all the cohesive elements on the interface Γ_* . The solution to Eq. (36) in the generalized case can be obtained an iterative solution procedure. Let ${}^{k+1}_{j+1}\mathbf{U}$ be the global (nodal) displacement vector at an applied load/displacement step $j+1$ and iteration $k+1$. Using a Taylor's expansion we can linearize the global displacement and the residual vectors as

$$\mathbf{R}({}^{k+1}_{j+1}\mathbf{U}) = \mathbf{0} = \mathbf{R}({}^k_{j+1}\mathbf{U} + {}^{k+1}_{j+1}\Delta\mathbf{U}) = \mathbf{R}({}^k_{j+1}\mathbf{U}) + \left[\frac{\partial \mathbf{R}({}^k_{j+1}\mathbf{U})}{\partial {}^k_{j+1}\mathbf{U}} \right] {}^{k+1}_{j+1}\Delta\mathbf{U}. \quad (37)$$

Rewriting the above equation into a fully discretized and linearized system of equation, we obtain

$${}^k_{j+1}\mathbf{K} {}^k_{j+1}\Delta\mathbf{U} = \mathbf{R}({}^k_{j+1}\mathbf{U}) \quad (38)$$

where $\mathbf{K}$ is the tangent matrix obtained by assembling the contributions of the bulk ($\mathbf{K}^b$) and cohesive ($\mathbf{K}^c$) tangent matrices as

$$\mathbf{K} = -\frac{\partial \mathbf{R}}{\partial \mathbf{U}} = \frac{\partial \mathbf{f}_{\text{int}}^b}{\partial \mathbf{U}} + \frac{\partial \mathbf{f}_{\text{int}}^c}{\partial \mathbf{U}} = \mathbf{K}^b + \mathbf{K}^c, \quad (39)$$

3.3. Bulk elements

The internal bulk force vector $\mathbf{f}_{\text{int}}^b$ is assembled as

$$\mathbf{f}_{\text{int}}^b = \sum_e \int_{\Omega_e^m} \mathbf{B}^T \boldsymbol{\sigma}^m = \sum_e \int_{\Omega_e^m} \mathbf{B}^T \mathbf{D}^m \mathbf{B} \bar{\mathbf{u}}^m d\Omega_e \quad \text{for } m = \{1, 2\}, \quad (40)$$

where $\sum_e$ indicates the matrix (or vector) assembly of the global system from the element matrices (or vectors) in the entire computational domain, $\mathbf{B}$ is the strain–displacement relationship matrix defined in Eq. (33), $\boldsymbol{\sigma}^m = [\sigma_{11}, \sigma_{22}, \sigma_{12}]_{3 \times 1}^T$ is the stress tensor in Voigt notation containing only the in-plane components, and $\mathbf{D}^m$ is the 3×3 plane strain elasticity matrix in Voigt notation. The corresponding bulk tangent matrix is

$$\mathbf{K}^b = \sum_e \int_{\Omega_e^m} \mathbf{B}^T \mathbf{D}^m \mathbf{B} d\Omega_e \quad \text{for } m = \{1, 2\}, \quad (41)$$

Neglecting body forces, $\mathbf{f}_{\text{ext}}$ is obtained by assembling the element contributions from any applied traction on the Neumann boundary

$$\mathbf{f}_{\text{ext}} = \sum_e \int_{\Gamma_{ne}} \mathbf{N}^T \bar{\mathbf{t}} d\Gamma_e \quad (42)$$

Note that Abaqus automatically handles the evaluation of $\mathbf{f}_{\text{ext}}$, so simple traction boundary conditions need not be defined using user defined subroutines.

3.4. Interface elements

The cohesive internal force vector $\mathbf{f}_{\text{int}}^c$ has contributions from both the consistency and stabilization terms in the variational form in Eq. (26) and can be assembled as

$$\mathbf{f}_{\text{int}}^c = \mathbf{f}^{\text{consistency}} + \mathbf{f}^{\text{stabilized}} = - \sum_e \int_{\Gamma_{*e}} \llbracket \mathbf{N} \rrbracket^T (\mathbf{I} - \mathbf{S}) \mathbf{T} \boldsymbol{\sigma}_\gamma d\Gamma_e + \sum_e \int_{\Gamma_{*e}} \llbracket \mathbf{N} \rrbracket^T \mathbf{S} \boldsymbol{\alpha} \delta d\Gamma_e, \quad (43)$$

where $\llbracket \mathbf{N} \rrbracket$ is the jump in shape function matrix in Eq. (35); $\mathbf{S}$, $\boldsymbol{\alpha}$, $\mathbf{I}$ are the 2×2 stabilization, cohesive stiffness and identity matrices, $\boldsymbol{\sigma}_\gamma = [\langle \sigma_{11} \rangle_\gamma, \langle \sigma_{22} \rangle_\gamma, \langle \sigma_{12} \rangle_\gamma]_{3 \times 1}^T$ is the weighted Cauchy stress for in-plane components in Voigt notation, and the 2×3 stress transformation matrix is

$$\mathbf{T} = \begin{bmatrix} -CS & CS & C^2 - S^2 \\ S^2 & C^2 & -2CS \end{bmatrix} \quad (44)$$

Thus, the matrix $\mathbf{T}$ defines the relation between the Cauchy stress tensor at any point on a bulk element edge and the traction vector at that point on the cohesive interface based on its orientation. The interface element's tangent stiffness matrix consists of both the consistency and stabilized terms and can be assembled as

$$\mathbf{K}^c = \mathbf{K}^{\text{consistency}} + \mathbf{K}^{\text{stabilized}} = \sum_e \int_{\Gamma_{*e}} \llbracket \mathbf{N} \rrbracket^T (\mathbf{I} - \mathbf{S}) \mathbf{T} \mathbf{D} \mathbf{B}_\gamma d\Gamma_e - \sum_e \int_{\Gamma_{*e}} \llbracket \mathbf{N} \rrbracket^T \mathbf{S} \boldsymbol{\alpha} \llbracket \mathbf{N} \rrbracket d\Gamma_e. \quad (45)$$

where $\mathbf{B}_\gamma$ denotes the weighted shape function gradient matrix defined as,

$$\mathbf{B}_\gamma = [\gamma^2 \mathbf{B}^2 \quad \gamma^1 \mathbf{B}^1]_{3 \times 16}. \quad (46)$$

In the above equation, $\mathbf{B}^1$ and $\mathbf{B}^2$ are matrices containing the gradient of the shape functions calculated from the adjacent bulk elements at the position of an interface Gauss point of the cohesive element. Note, by setting the matrix $\mathbf{S} = \mathbf{I}$ in Eq. (45) we can revert to the standard method.

3.5. Abaqus implementation

The proposed method is implemented in the commercial software Abaqus, as illustrated in Fig. 6. All the element force and stiffness matrices described in Sections 3.3 and 3.4 have been evaluated via the user-element-material (UEL MAT) and user-element (UEL) subroutines for 4-noded bulk and 4-noded interface elements, respectively.

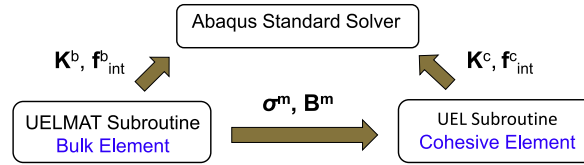


Fig. 6. Flow chart showing the interaction between UELMAT and UEL subroutines and the Abaqus standard solver for implementing the stabilized method through user-defined bulk and cohesive/interface elements.

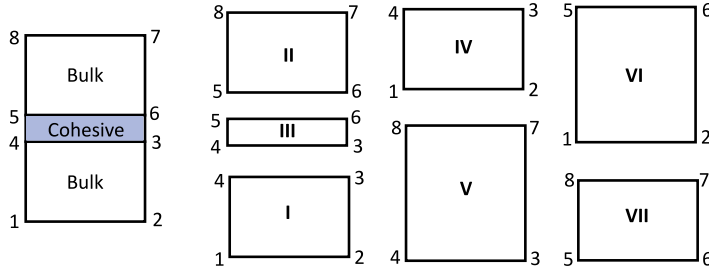


Fig. 7. Assembly of the cohesive element matrix $\mathbf{K}^{\text{consistency,e}}$ defined in Eq. (47) into the global tangent matrix in Abaqus requires the creation of four dummy elements (IV–VII) for each cohesive element (III).

The bulk element force vector $\mathbf{f}_{\text{int}}^b$ and the tangent matrix $\mathbf{K}^b$ are computed using the UELMAT subroutine, because it allows for the usage of in-built constitutive models via the `material.lib_mech` function. The cohesive element force vector $\mathbf{f}_{\text{int}}^c$ and the corresponding tangent matrix $\mathbf{K}^c$ are computed using the UEL subroutine, according to Eqs. (43) and (45). Recall that $\mathbf{f}_{\text{int}}^c$ and $\mathbf{K}^c$ contain the weighted average of stress σ_γ and shape function derivative matrix $\mathbf{B}_\gamma$, which are calculated using the nodal displacement vectors and shape function matrices of the two neighboring bulk elements. To avoid repetition of computations in our implementation, we calculate σ_γ and $\mathbf{B}_\gamma$ at the cohesive element Gauss points lying on the bulk element edges in the UELMAT subroutine and pass them to the UEL subroutine using global modules. The tangent matrix $\mathbf{K}^c$ is unsymmetric owing to the consistency term $\mathbf{K}^{\text{consistency}}$ in our formulation, whereas the stabilization term $\mathbf{K}^{\text{stabilized}}$ is symmetric. The matrix $\mathbf{B}_\gamma$ defined in Eq. (46) has the dimension of 8×16 , where the number of rows correspond to the interfacial degrees of freedoms (DoFs) and the number of columns correspond to the interfacial and adjacent bulk element DoFs (see Fig. 7). However, the UEL subroutine for the cohesive interface element allows access to only its four nodes (i.e., eight DoFs), so we can only assemble an 8×8 element stiffness matrix. To assemble the 8×16 element stiffness matrix computed in the UEL subroutine into the global stiffness matrix, we create “dummy” elements in the mesh (elements IV–VII in Fig. 7). We partition the 8×16 element stiffness matrix as,

$$[\mathbf{K}^{\text{consistency,e}}]_{8 \times 16} = \begin{bmatrix} [\mathbf{K}^{\text{IV}}]_{4 \times 4} & & [\mathbf{K}^{\text{V}}]_{4 \times 4} \\ & [\mathbf{K}^{\text{III}}]_{8 \times 8} & \\ [\mathbf{K}^{\text{VI}}]_{4 \times 4} & & [\mathbf{K}^{\text{VII}}]_{4 \times 4} \end{bmatrix}. \quad (47)$$

and assemble the 8×8 matrix through the cohesive element and the four 4×4 matrices through the dummy elements using global modules into the global stiffness matrix. Note that the dummy elements are only used for matrix assembly.

Although the above implementation with dummy elements for the cohesive stiffness matrix assembly may appear convoluted, it is efficient and may even be advantageous when interface elements are inserted along all element edges. This is because all bulk stress and shape function matrices at interface Gauss points are computed once in the UELMAT subroutine and stored in global modules, instead of recomputing it wherever needed in the UEL subroutine. We found that our implementation of the stabilized method did not increase the wall clock time of computation (compared to standard method) in all the numerical examples presented in Section 4. However, we

note that alternative implementations of the stabilized method in existing finite element codes are possible. For example, Versino et al. [65] used Abaqus UEL subroutine for a 8-noded interface element in 2D to implement a discontinuous Galerkin based extrinsic cohesive zone model. This implementation may be more advantageous when interface elements are inserted along simple (straight) interfaces between laminate plies in composite materials.

The UELMAT and UEL subroutines are detailed in Algorithms 1 and 2, respectively. As discussed in [61], we formulate the bilinear cohesive zone model within the damage mechanics framework that allows us to automatically handle the unloading/reloading conditions based on a previous maximum damage (history variable). Despite the bilinear shape of the cohesive law, the damage variable is a nonlinear function of interface separation and this nonlinearity is handled by Abaqus/Standard outside of the user subroutines. As detailed in the Abaqus manual [70, Chapter 7: Analysis Solution and Control], Abaqus/Standard combines incremental and iterative (Newton–Raphson) procedures for solving nonlinear problems. The total load/displacement is applied incrementally as smaller increments (pseudo-time steps) and the user typically suggests the size of the first increment and Abaqus/Standard automatically chooses the size of the subsequent increments. Within each increment, Abaqus/Standard automatically performs iteration to find an equilibrium solution based on a user-defined criteria for residual force and displacement correction. We note that, in cohesive fracture simulations, the Abaqus/Standard default criteria may be too small that numerical convergence may not be attainable. Therefore, we increase these tolerances appropriately so as to attain convergence and maintain adequate accuracy, to obtain the results in Sections 4.4 and 4.5.

Algorithm 1 : Abaqus UELMAT subroutine for the bulk element

Given all the variables at the previous iteration of the current increment, at the next iteration:

1. Compute the shape function derivative matrix $\mathbf{B}$ according to Eq. (33)
 2. Determine the 3×3 plane strain elasticity tensor $\mathbf{D}$ via the `material_lib_mech` function
 3. Compute and assemble the bulk element contributions to the tangent matrix and the internal force vector using four-point Gauss integration
 4. Determine σ^m and shape function derivatives $\mathbf{B}^m$ at the two interface Gauss integration points and store them in global modules
-

4. Numerical examples

In this section, we present four examples to demonstrate the numerical stability and accuracy of the proposed stabilized method in two-dimensions. For all the simulations we assumed bilinear quadrilateral plane strain elements with four-point Gauss integration scheme and four-noded linear cohesive elements with two-point Gauss integration scheme. Currently, the user element subroutines are written only for 2D plane-strain and plane-stress elements in Abaqus software. Additional patch tests and benchmark problems have been presented in [71,72].

4.1. Square plate with horizontal interface

In this example, we assess the accuracy of the stabilized method in recovering normal traction on a straight, horizontal interface with isotropic CZMs using a constant strain patch test [73]. We consider a square plate of side length $L = 1$ mm with a horizontal interface at mid-height (see Fig. 8a). Both vertical and horizontal displacements are constrained at the bottom edge of the plate, whereas a uniform vertical displacement $\Delta = 0.1$ mm is applied at the top edge of the plate, and traction-free conditions are specified on the left and right edges of the square plate. The Young's modulus and Poisson's ratio of the isotropic linearly elastic material in the bulk elements are assumed as $E = 1$ N/mm² and $\nu = 0.2$, respectively. We use a 10×10 structured square mesh with an element size of 0.1 mm and the stabilization parameters $\beta_n = \beta_\tau = 14$ N/mm³. The analysis is conducted under the assumption of small deformations and no interface damage (i.e., $d_s = 0$), although the bulk applied strain is 10%. The normal traction profile along the horizontal interface obtained from the standard and stabilized methods for different cohesive stiffness values is

Algorithm 2 : Abaqus UEL subroutine for the cohesive element

Given all the variables at the previous iteration of the current increment, at the next iteration:

1. Compute the jump in the interfacial shape function matrix $[\![\mathbf{N}]\!]$ according to Eq. (35) and the interface separation vector $[\![\mathbf{u}]\!]$ using Eq. (34)
2. Calculate equivalent separation δ_e , and equivalent critical δ_e^c and maximum δ^u separations according to Eq. (12), and static damage d_s using Eq. (10)
3. Calculate the stabilization matrix $\mathbf{S}$ according to Eq. (24)
4. Calculate the weighted stress $\langle \sigma \rangle_\gamma$ and shape function derivative matrix $\mathbf{B}_\gamma$, according to Eqs. (46) and using the information passed from the UELMAT subroutine
5. Compute and assemble the cohesive element contributions corresponding to the stabilized part of the tangent matrix and the internal force vector using two-point Gauss integration
6. Define the 3×3 plane strain elasticity tensor $\mathbf{D}$ and the stress transformation matrix $\mathbf{T}$.
7. Compute the cohesive element contributions corresponding to the consistency part of the tangent matrix and the internal force vector using two-point Gauss integration
8. Partition $[\mathbf{K}^{\text{consistency},e}]_{8 \times 16}$ into one 8×8 matrix $[\mathbf{K}^{\text{III}}]$ and four 4×4 matrices $[\mathbf{K}^{\text{IV}}]$, $[\mathbf{K}^{\text{V}}]$, $[\mathbf{K}^{\text{VI}}]$, $[\mathbf{K}^{\text{VII}}]$, as described in Eq. (47).
9. Assemble partitioned matrices into the global stiffness matrix using the cohesive element (III) and four dummy elements (IV–VII) (see Fig. 7).

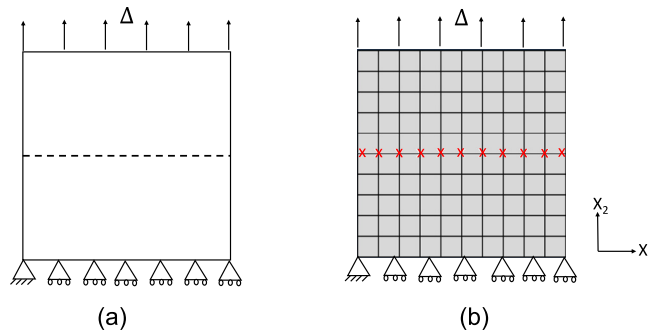


Fig. 8. Square plate with horizontal interface: (a) schematic diagram; (b) finite element mesh.

shown in Fig. 9. For the smaller cohesive stiffness value of 100 N/mm^3 both methods yield oscillation-free traction profile at the cohesive interface. For the larger stiffness value of 10^{16} N/mm^3 , the standard method exhibits instability resulting in spurious traction oscillations; whereas, the stabilized method does not exhibit any instability.

To demonstrate the accuracy of stabilized method for stiff cohesive laws, we report the relative l^2 -error (vector norm) in normal traction and separation for different values of initial cohesive stiffness in Table 1. The relative l^2 -errors are calculated as

$$\frac{\|\varepsilon_{t_n}\|_2}{\|t_n^*\|_2} = \frac{\sqrt{\sum_{i=1}^{N_{\text{GP}}} (t_n^i - t_n^{*i})^2}}{\sqrt{\sum_{i=1}^{N_{\text{GP}}} (t_n^{*i})^2}}; \quad \frac{\|\delta_n\|_2}{\|u_n^*\|_2} = \frac{\sqrt{\sum_{i=1}^{N_{\text{GP}}} (\delta_n^i)^2}}{\sqrt{\sum_{i=1}^{N_{\text{GP}}} (u_n^{*i})^2}} \quad (48)$$

where N_{GP} is the total number of Gauss (integration) points on the cohesive interface, interface separation δ_n and traction t_n are evaluated using Eqs. (6) and (29) respectively, and the reference values of the traction t_n^* and

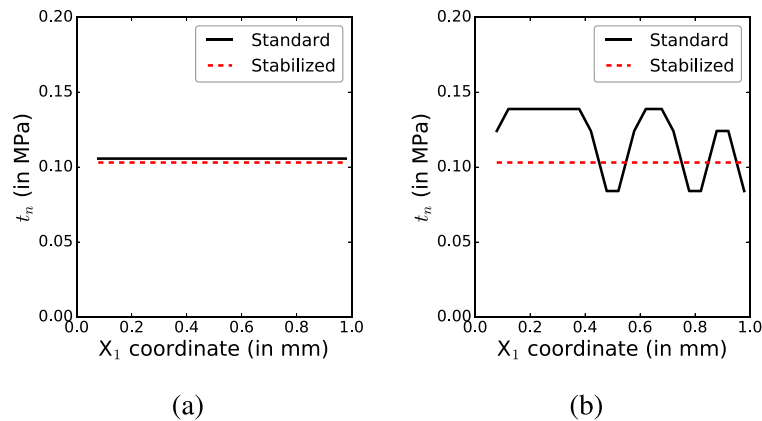


Fig. 9. Square plate with a horizontal interface: normal traction profiles obtained from the standard and stabilized methods with (a) $\alpha_n^0 = \alpha_\tau^0 = 10^2 \text{ N/mm}^3$ and (b) $\alpha_n^0 = \alpha_\tau^0 = 10^{16} \text{ N/mm}^3$.

Table 1

Square plate with a horizontal cohesive interface: relative l^2 -error in normal traction and separations from the standard and stabilized finite element methods for different cohesive stiffness. The bilinear isotropic CZM is used, wherein the tangential and normal cohesive stiffness are taken to be equal.

Cohesive stiffness (N/mm^3)	Standard FEM		Stabilized FEM	
	$\frac{\ e_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \delta_n\ _2}{\ u_n^*\ _2}$	$\frac{\ e_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \delta_n\ _2}{\ u_n^*\ _2}$
10^2	1.4%	2.1×10^{-2}	1%	2.1×10^{-2}
10^8	2.5%	2.2×10^{-8}	1×10^{-8}	2×10^{-8}
10^{15}	4.6%	2.2×10^{-15}	3.7×10^{-15}	2×10^{-15}
10^{16}	25.7%	1.3×10^{-16}	1.8×10^{-17}	2.4×10^{-16}

displacement u_n^* at the horizontal interface are calculated for the perfectly bonded interface case under the plane strain linear elastic assumption as

$$t_n^* = \left(\frac{\Delta}{L} \right) \frac{E}{(1 - \nu^2)}; \quad u_n^* = \frac{\Delta}{2}. \quad (49)$$

In the above equation, the normal traction is calculated based on the engineering strain (defined with respect to the initial length) under the assumption of small deformations and $E/(1 - \nu^2)$ is the plane strain elastic modulus [74]. From Table 1 it is evident that the stabilized method ensures accurate recovery of interface traction compared to the standard method. As the cohesive stiffness is increased to a large value the interface separation tends to zero (to machine precision) in both the standard and stabilized methods, but the error in interface traction tends to zero (to machine precision) only in the stabilized method. We also note that the computational time with the standard and stabilized method is comparable. We next investigate the effect of mesh refinement on the accuracy of traction evaluation for structured meshes. For an initial cohesive stiffness of 10^8 N/mm^3 , we see that the traction error does not change with mesh refinement in both methods, as given in Table 2. This further illustrates that the improvement in accuracy with the stabilized method ensues from the consistent weak formulation in Eq. (26).

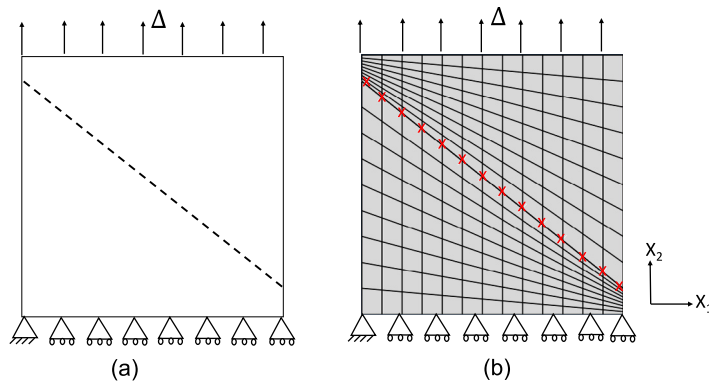
4.2. Square plate with inclined interface

In this example, we assess the accuracy of the stabilized method in recovering normal and tangential tractions on a straight, inclined interface with isotropic and anisotropic CZMs using the constant strain patch test [73]. We consider a square plate of side length $L = 1 \text{ mm}$ with a straight interface inclined at an initial angle of 140.4° with the global x_1 (i.e., horizontal) axis, as shown in Fig. 10. We use a 13×18 semi-structured mesh so that the interface is divided into 13 elements. The support conditions, loading and material properties are identical to

Table 2

Square plate with a horizontal cohesive interface: relative l^2 -error in normal traction and separations from the standard and stabilized finite element methods for different mesh resolutions. The bilinear isotropic CZM is used, wherein the tangential and normal cohesive stiffness are taken to be 10^8 N/mm³.

Mesh size (mm)	Standard FEM		Stabilized FEM	
	$\ \varepsilon_{tn}\ _2$	$\ \delta_n\ _2$	$\ \varepsilon_{tn}\ _2$	$\ \delta_n\ _2$
	$\ t_n^*\ _2$	$\ u_n^*\ _2$	$\ t_n^*\ _2$	$\ u_n^*\ _2$
0.1	2.5%	2.2×10^{-8}	1×10^{-8}	2×10^{-8}
0.04	2.5%	2.2×10^{-8}	1×10^{-8}	2×10^{-8}
0.02	2.5%	2.2×10^{-8}	1×10^{-8}	2×10^{-8}

**Fig. 10.** Square plate with inclined interface: (a) schematic diagram; (b) finite element mesh.

those discussed in Section 4.1, and the stabilization parameters $\beta_n = \beta_\tau = 30$ N/mm³. The analysis is conducted under the assumption of small deformations and no interface damage (i.e., $d_s = 0$). The normal and tangential tractions along the inclined interface obtained from the standard and stabilized methods for different cohesive stiffness values are shown in Fig. 11. For the smaller cohesive stiffness value of 100 N/mm³ both methods yield oscillation-free traction profiles at the cohesive interface. For the larger stiffness value of 10^{16} N/mm³, the standard method exhibits instability resulting in spurious traction oscillations; whereas, the stabilized method does not exhibit such an instability.

To illustrate the accuracy of the stabilized method for stiff isotropic CZMs, we report the relative l^2 -error in normal and tangential tractions for different values of initial cohesive stiffness using the isotropic bilinear CZM in Table 3. The reference value of the normal and tangential tractions t_n^* and t_τ^* under the plane strain and perfectly bonded interface assumptions can be evaluated as

$$t_n^* = \left(\frac{\Delta}{L}\right) \frac{E}{(1-\nu^2)} \sin^2 \theta; \quad t_\tau^* = \left(\frac{\Delta}{L}\right) \frac{E}{(1-\nu^2)} \sin \theta \cos \theta, \quad (50)$$

where $\theta = 140.4^\circ$ is the angle that the inclined interface makes with the global x_1 axis in the undeformed (initial) configuration. From Table 3, it is evident that the stabilized method ensures accurate recovery of interface traction for large values of cohesive stiffness; whereas, in the standard method the interface traction error increases with the cohesive stiffness.

We next examine the performance of the standard and stabilized methods for anisotropic CZMs, wherein the normal and tangential cohesive stiffness values are not equal. It can be seen from Fig. 12, that the standard method yields an oscillation-free traction profile for smaller stiffness values, but it suffers from instability for larger values with spurious oscillations in the normal traction profile. In contrast, the stabilized method yields an oscillation-free traction profile regardless of the choice of cohesive stiffness. In Table 4 we report the accuracy of the stabilized and standard methods for the anisotropic CZM. We observe that for the assumed normal and tangential stiffness values, the stabilized method recovers crack-surface traction more accurately compared to the standard method; the

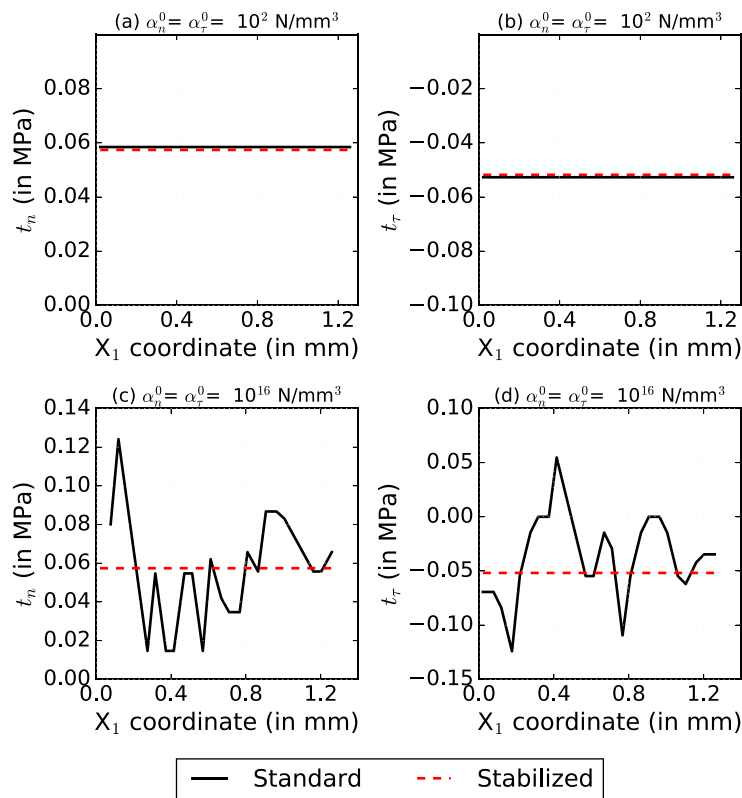


Fig. 11. Square plate with an inclined interface: traction profiles obtained from the standard and stabilized methods for isotropic cohesive zone models for two different cohesive stiffness values.

Table 3

Square plate with inclined interface: relative l^2 -error in normal and tangential tractions from the standard and stabilized finite element methods for different cohesive stiffness. The bilinear isotropic CZM is used, wherein the tangential and normal cohesive stiffness are taken to be equal.

Cohesive stiffness (N/mm ³)	Standard FEM		Stabilized FEM	
	$\frac{\ \varepsilon_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \varepsilon_{t_\tau}\ _2}{\ t_\tau^*\ _2}$	$\frac{\ \varepsilon_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \varepsilon_{t_\tau}\ _2}{\ t_\tau^*\ _2}$
10^2	1.7%	1.8%	0.7%	0.8%
10^8	2.6%	2.6%	1.1×10^{-7}	3.4×10^{-8}
10^{14}	3.0%	2.6%	8.3×10^{-10}	1.1×10^{-10}
10^{16}	64.9%	80.2%	9.3×10^{-11}	9.1×10^{-11}

error in traction decreases close to machine precision in the stabilized method as the stiffness is increased to a very large value.

4.3. Square plate with semicircular interface

In the previous examples, we considered constant strain patch tests with straight interfaces described by isotropic and anisotropic CZMs; in these cases traction oscillations do not appear if full Gauss integration is used with the standard method, except if the initial cohesive stiffness values are assumed to be very large (i.e., $\alpha_n^0 \geq 10^{15} \text{ N/mm}^3$). However, Svenning [31] has shown that spurious tractions oscillations can appear with curved interfaces described by an anisotropic CZM under compression, even for moderately large values of cohesive stiffness (e.g., 2–4 orders of magnitude more than the Young's modulus). Therefore, in this example, we demonstrate the ability of the stabilized

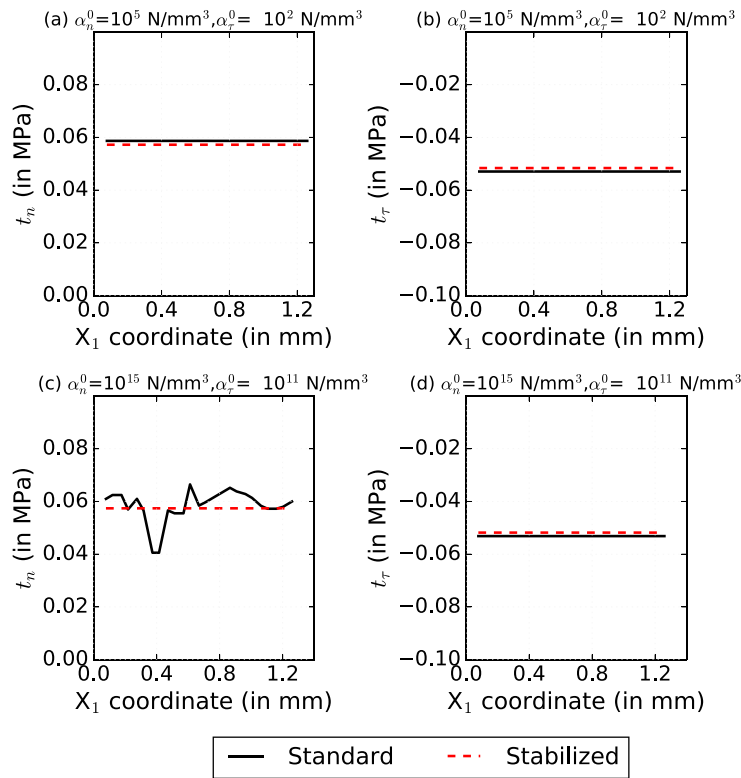


Fig. 12. Square plate with an inclined interface: traction profiles obtained from the standard and stabilized methods for anisotropic cohesive zone models for different cohesive stiffnesses.

Table 4

Square plate with inclined interface: relative l^2 -error in normal and tangential tractions from the standard and stabilized finite element methods for different cohesive stiffness. The bilinear anisotropic CZM is used, wherein the tangential and normal cohesive stiffness are taken to be different from each other.

Cohesive stiffness (N/mm ³)	Standard FEM		Stabilized FEM	
	$\frac{\ \varepsilon_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \varepsilon_{t_\tau}\ _2}{\ t_\tau^*\ _2}$	$\frac{\ \varepsilon_{t_n}\ _2}{\ t_n^*\ _2}$	$\frac{\ \varepsilon_{t_\tau}\ _2}{\ t_\tau^*\ _2}$
$\alpha_n^0 = 10^5, \alpha_\tau^0 = 10^2$	2.2%	2.2%	0.3%	0.3%
$\alpha_n^0 = 10^{11}, \alpha_\tau^0 = 10^7$	2.6%	2.6%	1.1×10^{-7}	4.9×10^{-8}
$\alpha_n^0 = 10^{15}, \alpha_\tau^0 = 10^{11}$	10.5%	2.6%	8.3×10^{-10}	1.1×10^{-10}

method to alleviate traction oscillations on a semicircular interface with an anisotropic CZM. We consider a square plate of side length $L = 100$ mm with a cohesive interface of diameter 60 mm, as shown in Fig. 13. A horizontal compressive displacement of $\Delta = 1$ mm is applied on the right edge of the plate, the displacement at the left edge is constrained in both directions, and traction-free condition is specified for the top and bottom edges of the plate. We assume the plane strain condition and a linear elastic bulk material with a modulus of elasticity $E = 20$ GPa, and a Poisson's ratio $\nu = 0.2$, following the example in [31]. The analysis is conducted under the assumption of small deformations and no interface damage (i.e., $d_s = 0$).

We first examine the performance of the standard method for isotropic and anisotropic CZMs using unstructured meshes. We consider a coarse mesh with 240 elements and a fine mesh with 1780 elements with bilinear quadrilateral elements. For the isotropic CZM we take normal and tangential cohesive stiffnesses $\alpha_n^0 = \alpha_\tau^0 = 10^{11}$ N/mm³ and for the anisotropic CZM we take $\alpha_n^0 = 10^{11}$ N/mm³ and $\alpha_\tau^0 = 10^9$ N/mm³. Fig. 14 shows the normal traction profiles

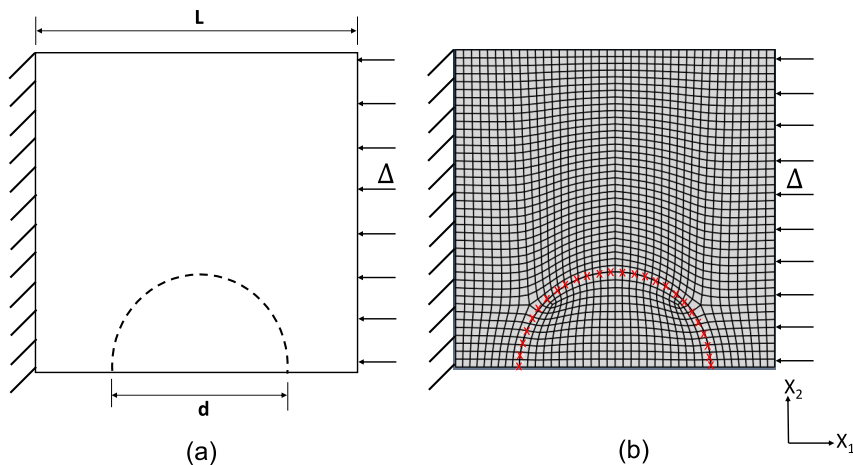


Fig. 13. Square plate with semicircular interface: (a) schematic diagram; (b) finite element mesh.

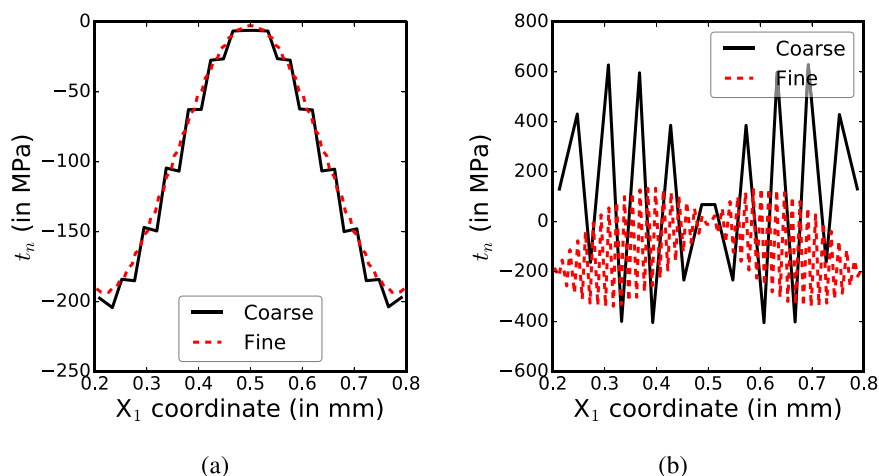


Fig. 14. Square plate with semicircular interface: effect of mesh refinement on normal traction profiles predicted by the standard method. (a) $\alpha_n^0 = \alpha_\tau^0 = 10^{11}$ N/mm³ and (b) $\alpha_n^0 = 10^{11}$ N/mm³ and $\alpha_\tau^0 = 10^9$ N/mm³.

along the cohesive interface obtained from the standard method. Fig. 14(a) indicates that the standard method yields a smooth traction profile with the isotropic CZM if the mesh is adequately refined. However, mesh refinement cannot alleviate spurious traction oscillations with the anisotropic CZM, as evident from Fig. 14(b). This issue with the standard (penalty-like) method for anisotropic CZMs was previously reported in [31].

In Fig. 15, we compare the performance of the standard and stabilized methods for isotropic and anisotropic CZMs using the fine mesh (with 1780 elements). According to Eq. (25), we take the stabilization parameters $\beta_n = \beta_\tau = 2 \times 10^5$ N/mm³. Fig. 15(a) and (b) show that for isotropic CZMs the standard method suffers from numerical instability only for very large values of cohesive stiffness (i.e., 10^{18} N/mm³); whereas, the stabilized method yields an oscillation-free traction profile. Fig. 15(c) shows that the stabilized method is able to alleviate the spurious oscillations in normal traction observed with the standard method for the anisotropic CZM with $\alpha_n^0 = 10^{11}$ N/mm³ and $\alpha_\tau^0 = 10^9$ N/mm³. These results illustrate that the standard method is not robust when dealing with stiff anisotropic cohesive laws, unlike the stabilized method.

We next evaluate the relative l^2 -error in interface traction from the standard and stabilized methods. Assuming that the interface is perfectly bonded for large values of cohesive stiffness $\alpha_n^0 > 10^8$ N/mm³, $\alpha_\tau^0 > 10^6$ N/mm³, the exact value of the normal and tangential tractions t_n^* and t_τ^* at interface Gauss points can be calculated using Eq. (50), wherein θ is the local orientation of the interface element that varies along semicircular interface. Table 5

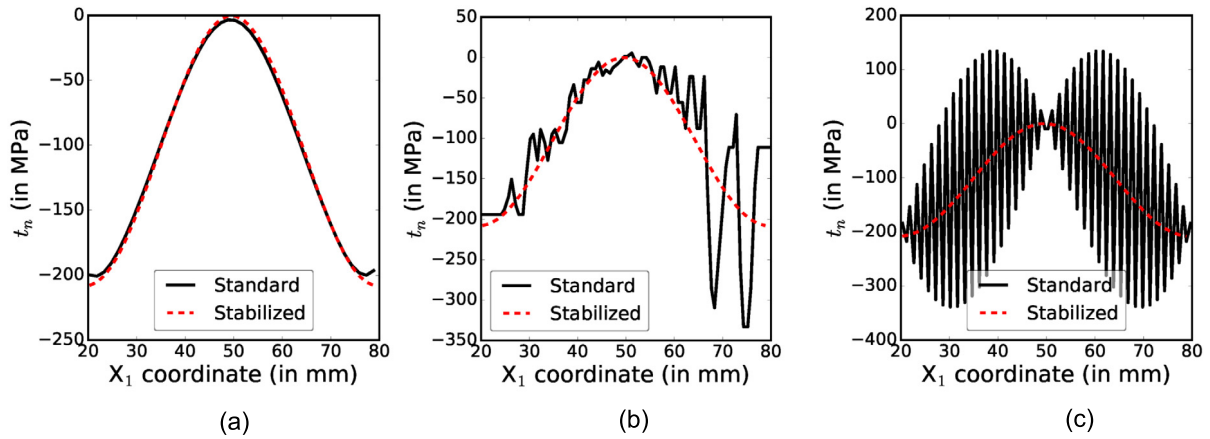


Fig. 15. Square plate with semicircular interface: Normal traction profiles obtained from the standard and stabilized method for different cohesive stiffness. (a) isotropic CZM with $\alpha_n^0 = \alpha_\tau^0 = 10^{11}$ N/mm³, (b) isotropic CZM $\alpha_n^0 = \alpha_\tau^0 = 10^{18}$ N/mm³ and (c) anisotropic CZM $\alpha_n^0 = 10^{11}$ N/mm³ and $\alpha_\tau^0 = 10^9$ N/mm³.

shows that the stabilized method ensures the accurate recovery of normal traction at the interface for both isotropic and anisotropic CZMs, unlike the standard method. Even if the cohesive stiffness $\alpha_n^0 = 10^8$ N/mm³ and $\alpha_\tau^0 = 10^6$ N/mm³ are 2–4 orders of magnitude larger than the Young's modulus $E = 20 \times 10^3$ N/mm³, we see that the standard method performs quite poorly with 117% error in normal traction and exhibits spurious oscillations (results are identical to Fig. 14b). Noting that in mode II fracture analysis, an anisotropic cohesive law with $\alpha_n^0 > \alpha_\tau^0$ is typically used to enforce the no inter-penetration condition at the crack surface, the stabilized method can offer a significant advantage owing to its stability and accuracy.

4.4. Asymmetric double cantilever beam

In all the previous examples, we conducted linear elastic analysis under the assumption of small deformations and no interface damage (i.e., $d_s = 0$) to illustrate the accuracy of the stabilized method in recovering crack-surface traction. In this example, we will examine the accuracy of the stabilized method for analyzing mixed-mode delamination crack growth in composite materials using the asymmetric double cantilever beam. While the specimen geometry and test set-up shown in Fig. 16 resembles that in [75,76], we altered the applied load configuration. The fixed boundary condition is applied at the right end of the beam, and vertical displacements Δ_1 and Δ_2 are applied at the upper and lower nodes at the left end (with $\Delta_2/\Delta_1 = 0.095$), to initiate the delamination process. Thus, using a displacement-controlled simulation, we capture the softening portion of the load–displacement curve due to the evolution of damage d_s in the interface elements; however, the mode-mix ratio between mode I and mode II fracture is not a constant, because the applied loads are not constant. We perform all simulation studies using a structured mesh with rectangular bulk elements (0.125 mm $\times$ 1 mm) and interface element size of 0.125 mm, so that the cohesive process zone is adequately resolved according to the guidelines described in [77]. The material properties and cohesive parameters assumed for this test are listed in Table 6, and the stabilization parameters $\beta_n = \beta_\tau = 2 \times 10^6$ N/mm³.

To solve this nonlinear quasi-static fracture problem, we need to linearize the discretized equilibrium equation using the Taylor series expansion (see Eq. (37)) at a given iteration and applied load/displacement step; in order to ensure accuracy it may be necessary to take the applied displacement increment small enough within a load/displacement step [79]. In Abaqus, this can be achieved by prescribing a small displacement increment $\dot{\Delta}$ within the default quasi-static pseudo-time stepping algorithm. To avoid convergence issues with nonlinear analysis in Abaqus, we choose the force residual tolerance $R_n^\alpha = 1.0$ [70, Chapter 7: Analysis Solution and Control]. Because the weak form and the corresponding discretized equilibrium equation from standard and stabilized methods are different, we investigate their accuracy in relation to the choice of displacement increment along with the cohesive

Table 5

Square plate with semicircular interface: accuracy of the standard and stabilized finite element methods. We consider both isotropic and anisotropic CZMs and relative l^2 -error in normal tractions is reported.

Cohesive stiffness (N/mm ³)	Standard FEM $\frac{\ \varepsilon_n\ _2}{\ t_n^*\ _2}$	Stabilized FEM $\frac{\ \varepsilon_n\ _2}{\ t_n^*\ _2}$
$\alpha_n^0 = \alpha_\tau^0 = 10^8$	2.9%	4.3×10^{-8}
$\alpha_n^0 = \alpha_\tau^0 = 10^{11}$	2.9%	4.3×10^{-8}
$\alpha_n^0 = 10^8, \alpha_\tau^0 = 10^6$	117.1%	4.2×10^{-8}
$\alpha_n^0 = 10^{11}, \alpha_\tau^0 = 10^9$	117.6%	1.3×10^{-7}

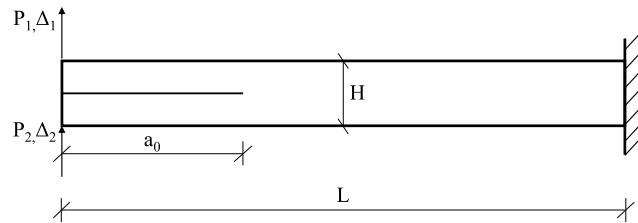


Fig. 16. Geometry and boundary conditions for the asymmetric double cantilever beam test. The dimensions are: $L = 100$ mm, $H = 4$ mm and $a_0 = 25$ mm.

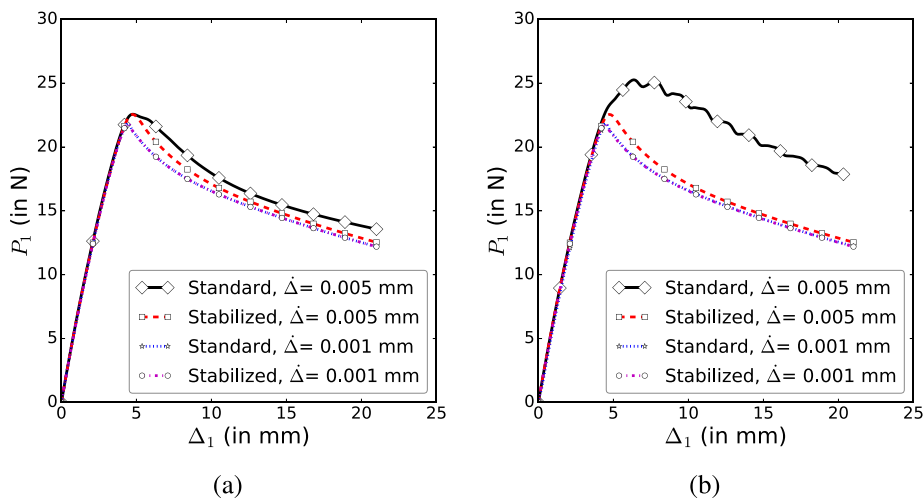


Fig. 17. Load versus displacement curves for the asymmetric double cantilever beam test: (a) $\alpha_n^0 = \alpha_\tau^0 = 10^8$ N/mm³ and (b) $\alpha_n^0 = \alpha_\tau^0 = 10^{12}$ N/mm³.

Table 6

Material properties and cohesive parameters for the asymmetric double cantilever beam. The cohesive parameters are assumed from [78] for mixed-mode fracture.

E (N/mm ²)	ν	G_{IC} (N/mm)	G_{IIC} (N/mm)	σ_{max} (N/mm ²)	τ_{max} (N/mm ²)
10^5	0.35	4.0	4.0	57	57

stiffness. In Fig. 17, we plot the load–displacement responses for two different displacement increments (0.005 mm/step and 0.001 mm/step) and cohesive stiffness values ($\alpha_n^0 = \alpha_\tau^0 = 10^8$ N/mm³ and $\alpha_n^0 = \alpha_\tau^0 = 10^{12}$ N/mm³). For the smaller stiffness of 10^8 N/mm³ (see Fig. 17a), the standard and stabilized methods predict the same load–displacement curve for $\dot{\Delta} = 0.001$ mm/step; however, both methods are slightly inaccurate for $\dot{\Delta} = 0.005$ mm/step.

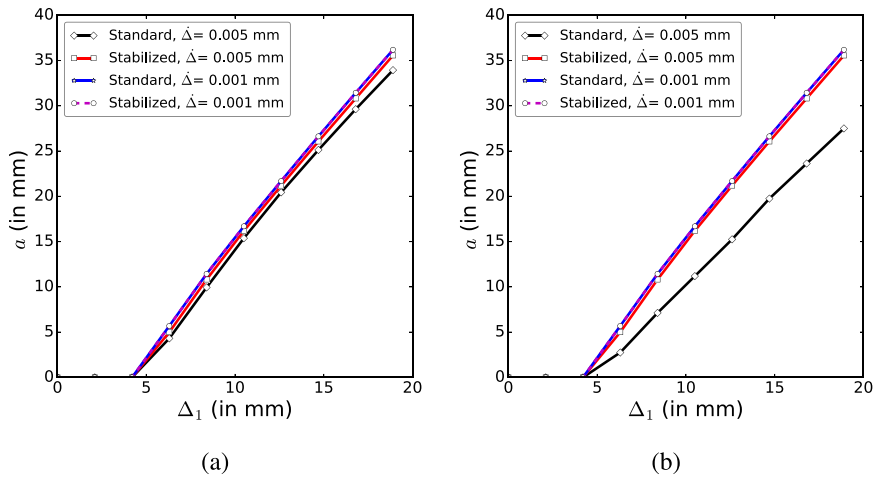


Fig. 18. Crack versus displacement curves for the asymmetric double cantilever beam test: (a) $\alpha_n^0 = \alpha_\tau^0 = 10^8 \text{ N/mm}^3$ and (b) $\alpha_n^0 = \alpha_\tau^0 = 10^{12} \text{ N/mm}^3$.

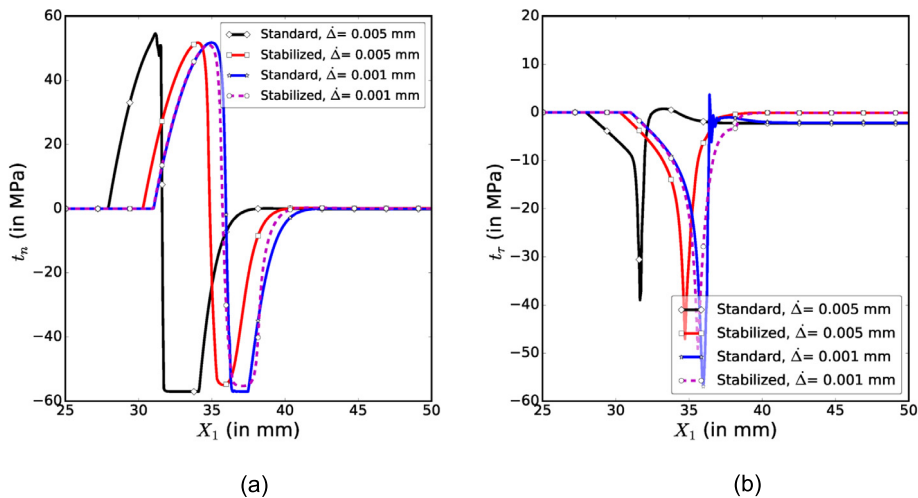


Fig. 19. Traction versus interface length curves for the asymmetric double cantilever beam test for $\alpha_n^0 = \alpha_\tau^0 = 10^{12} \text{ N/mm}^3$: (a) normal traction and (b) tangential traction.

We also used a smaller displacement increment $\dot{\Delta} = 0.0005 \text{ mm/step}$ (results not shown) and observed that the load–displacement curves match exactly with those obtained with $\dot{\Delta} = 0.001 \text{ mm/step}$. For the larger stiffness of 10^{12} N/mm^3 (see Fig. 17b), the standard and stabilized methods predict the same load–displacement curve for $\dot{\Delta} = 0.001 \text{ mm/step}$, but the standard method is significantly inaccurate for $\dot{\Delta} = 0.005 \text{ mm/step}$ and shows oscillations in the softening portion of the load–displacement curve. In contrast, the load–displacement curve obtained from the stabilized method for $\dot{\Delta} = 0.005 \text{ mm/step}$ is reasonably accurate without any oscillations in the softening portion.

To further explore the reason behind the inaccuracy in load-predictions for larger displacement rates, we plot the crack versus displacement curves in Fig. 18 for two cohesive stiffness values considered above. For the smaller stiffness of 10^8 N/mm^3 (see Fig. 18a), the standard and stabilized methods predict the same crack growth behavior for the two displacement rates. However, for the larger stiffness of 10^{12} N/mm^3 (see Fig. 18b), the standard method predicts slower crack growth for $\dot{\Delta} = 0.005 \text{ mm/step}$, which leads to the inaccurate prediction of softening portion of the load–displacement response. In contrast, the stabilized method predicts reasonably similar crack growth behavior for the two displacement rates. We next examine the normal and tangential traction profiles along the cohesive interface predicted by the standard and stabilized methods for the larger cohesive stiffness of 10^{12} N/mm^3 .

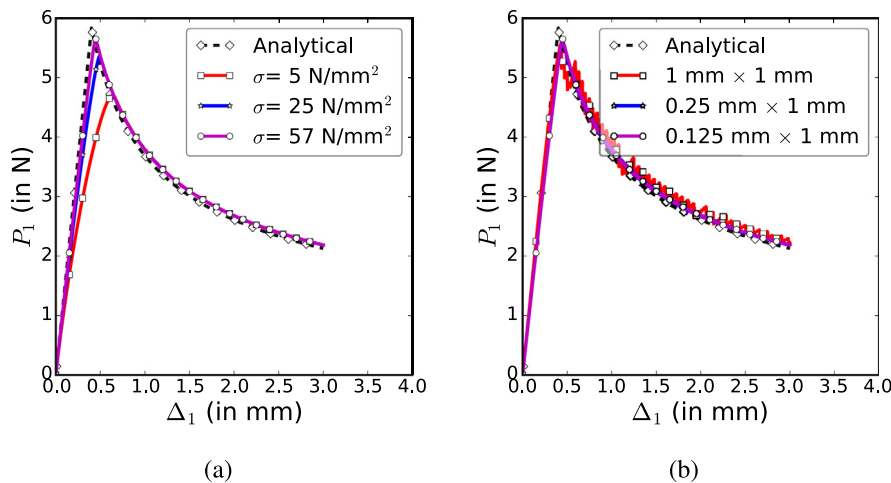


Fig. 20. Load versus displacement curves for the symmetric double cantilever beam test: (a) different cohesive interface strength and (b) different mesh size.

Table 7

Material properties and model parameters for the double cantilever beam. The cohesive parameters are assumed from [78] for mode I fracture.

E (N/mm ²)	ν	G_{IC} (N/mm)	σ_{max} (N/mm ²)
10^5	0.35	0.28	57

In Fig. 19, we plot the traction versus interface length at an applied displacement $\Delta = 6.4$ mm for $\dot{\Delta} = 0.005$ mm/step and 0.001 mm/step. In Fig. 19(a), the normal traction profiles from both methods match well for $\dot{\Delta} = 0.001$ mm/step; but the profile predicted by the standard method for $\dot{\Delta} = 0.005$ mm/step shows traction oscillations near the tension peak and plateau region near the compression peak. Because the crack length predicted by the standard method is smaller, the corresponding traction profile lags behind the other profiles. The tangential traction profiles in Fig. 19(b) also show similar behavior. Notably, oscillations in tangential traction can be observed even for $\dot{\Delta} = 0.001$ mm/step in the standard method. In summary, this study illustrates that the accurate recovery of crack-surface traction by the stabilized method can mitigate spurious sensitivity of load–displacement curves to displacement increment, and thus enable the reliable prediction of delamination crack propagation.

4.5. Double cantilever beam

In this example, we investigate the accuracy of the stabilized method in recovering crack-face traction for mode-I delamination crack growth using the double cantilever beam (DCB). We also examine the sensitivity of load–displacement curves to interface cohesive strength and mesh/element size. The specimen geometry and test set-up are identical to that in the previous example shown in Fig. 16, except for the applied load configuration. Specifically, at the left end of the beam we apply vertical displacements Δ_1 and Δ_2 on the upper and lower nodes with $\Delta_1/\Delta_2 = -1$ to initiate the delamination process. Thus, using a displacement-controlled simulation, we capture the softening portion of the load–displacement curve due to the evolution of damage d_s in the interface elements. The material properties and cohesive parameters assumed for this test are listed in Table 7. We choose the displacement increment $\dot{\Delta} = 0.0001$ mm/step, cohesive stiffness $\alpha_n^0 = 10^8$ N/mm³, and the stabilization parameters $\beta_n = \beta_\tau = 2 \times 10^6$ N/mm³ for all the simulations.

In Fig. 20, we plot the load–displacement responses for different interface strengths and mesh sizes along with the linear elastic analytical solution given in [78]. For the fine mesh with 0.125 mm $\times$ 1 mm rectangular elements, Fig. 20a shows that the cohesive strength effects the peak load prediction, and it is important to take the cohesive strength adequately large to ensure a better match with the analytical solution. For smaller cohesive strengths, the reduced peak load prediction is a consequence of crack initiation and propagation at smaller applied displacements.

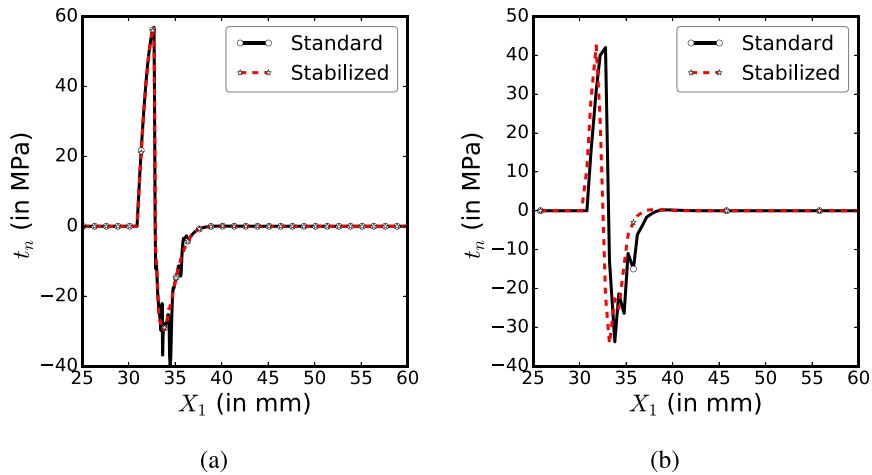


Fig. 21. Normal traction versus interface length curves from the standard and stabilized methods for the double cantilever beam test for $\alpha_n^0 = \alpha_\tau^0 = 10^8$ N/mm³: (a) fine mesh ($h = 0.125$ mm) and (b) coarse mesh ($h = 1$ mm).

Beyond a certain value of the interface strength ($\sigma_{max} = 57$ N/mm²) there is little effect from increasing the cohesive strength, as the load–displacement curve converges to the analytical solution. However, increasing the cohesive strength decreases the cohesive process zone size, so a smaller element size is required to accurately recover the crack-face traction. Fig. 20b depicts the effect of mesh size on the load–displacement response for cohesive strength $\sigma_{max} = 57$ N/mm². Choosing a coarse mesh (e.g., $h = 1$ mm) yields a noisy load–displacement curve due to inaccuracies in crack-face traction and episodic crack growth. Our study suggests that the interface element size $h = \text{meas}(\Gamma_*)$ should be chosen smaller than 0.25 mm to better capture the softening portion of the load–displacement curve. This is consistent with the guidelines described in [77] that the cohesive process zone needs to be resolved with at least three interface elements to ensure sufficient accuracy.

We next compare the normal traction profile along the cohesive interface obtained from the standard and the stabilized method at an applied displacement $\Delta = 0.67$ mm for coarse and fine meshes. In Fig. 21, the traction profile obtained from the standard method shows spurious oscillations, whereas that obtained from the stabilized method is free of oscillations. We also observe that the peak traction in the coarse mesh ($h = 1$ mm) is less than the cohesive strength $\sigma_{max} = 57$ N/mm², whereas that in the fine mesh ($h = 0.125$ mm) it is equal to the cohesive strength, which illustrates that a finer mesh is required to accurately capture the crack-face traction. Thus, this study demonstrates the superior stability of the stabilized method compared to the standard method for coarse and fine meshes.

5. Conclusion

In this paper, we proposed a stabilized finite element method for enforcing stiff isotropic and anisotropic cohesive laws using zero-thickness interface elements. The stabilized method generalizes Nitsche’s method to cohesive fracture problems and the key advantage is that our method remains well defined for any arbitrarily large value of cohesive stiffness. We presented several numerical examples demonstrating the stability and accuracy of the proposed method over the standard (penalty-like) method in two-dimensions. We first determined the numerical accuracy of the stabilized method in recovering crack surface traction at straight and semi-circular interfaces using constant strain patch tests. We demonstrated that the traction error in the stabilized method (measured with respect to the analytical solution for perfectly-bonded interface) approaches machine precision for large values of cohesive stiffness; whereas, the error increased in the standard method as the cohesive stiffness was increased. We next evaluated the numerical stability of the proposed method in alleviating spurious traction oscillations along the interface for stiff isotropic and anisotropic cohesive laws (i.e., with equal and unequal normal and tangential stiffnesses). Our numerical results clearly showed the presence of spurious traction oscillations in the standard method when enforcing anisotropic cohesive laws on curved interfaces under compression and sliding; whereas, the

stabilized method yielded oscillation-free traction profiles and ensured accurate recovery of crack surface traction, regardless of the choice of cohesive stiffness.

We next simulated mixed-mode delamination crack growth in an isotropic material using the asymmetric double cantilever beam test configuration. For stiff isotropic cohesive laws, we investigated the sensitivity of load–displacement curves predicted by the standard and stabilized methods to applied displacement increment (or load step size) and cohesive stiffness. If the displacement increment is taken small enough ($\dot{\Delta} = 0.001$ mm/step), then both the standard and stabilized methods predict the same load–displacement curve. However, for the larger cohesive stiffness and displacement increment of $\dot{\Delta} = 0.005$ mm/step, the standard method is less accurate compared to the stabilized method (see Fig. 17). Our results indicate that this discrepancy in the standard method potentially arises from the inaccurate prediction of crack growth behavior (see Fig. 18) and crack surface traction (see Fig. 19). In contrast, the stabilized method is sufficiently accurate in predicting peak load and crack growth even for the larger displacement increment $\dot{\Delta} = 0.005$ mm/step. Thus, this study illustrated that stabilized method could mitigate the spurious sensitivity of load–displacement results to displacement increment in mixed-mode fracture simulation. Finally, we investigated the effect of cohesive interface strength and mesh refinement on the load–displacement response using the mode I double cantilever beam test. We illustrated that the cohesive strength parameter affects the peak-load prediction and choosing a larger value ($\sigma_{max} = 57$ N/mm²) ensures a better match with the linear elastic analytical solution obtained from beam theory. We also show that it is necessary to choose the interface element size small enough (according to the criteria in [26,77]), in order to obtain accurate load–displacement curve and crack-face traction profiles. These cohesive fracture simulation studies clearly illustrate the superior stability and accuracy of the proposed Nitsche-based, stabilized finite element method compared to the standard finite element method.

Acknowledgments

GG and RD gratefully acknowledge the financial support of the United States Office of Naval Research – award #N0014-17-12040 (Program Officer: Mr. William Nickerson). We also thank Prof. Caglar Oskay at Vanderbilt University for his helpful comments on composite delamination analysis.

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